

BLOCKS

Types of Blocks in Java

1) Static Block

- Declared using `static { }`
- Executes **only once** when the class is loaded
- Used for **static initialization**

2) Non-Static Block (Instance Block)

- Declared using `{ }` (without static)
- Executes **every time an object is created**
- Runs **before constructor**

◆ Example of Static Block

```
class Test {  
    static {  
        System.out.println("Static block executed");  
    }  
  
    public static void main(String[] args) {  
        System.out.println("Main method");  
    }  
}
```

Output:

```
Static block executed  
Main method
```

◆ Example of Non-Static Block

```
class Test {  
    {  
        System.out.println("Non-static block executed");  
    }  
  
    Test() {  
        System.out.println("Constructor executed");  
    }  
  
    public static void main(String[] args) {  
        Test t1 = new Test();  
    }  
}
```

Output:

Non-static block executed
Constructor executed

◆ One Program with Both Blocks

```
class Test {  
  
    static {  
        System.out.println("Static block");  
    }  
  
    {  
        System.out.println("Non-static block");  
    }  
  
    Test() {  
        System.out.println("Constructor");  
    }  
}
```

```
public static void main(String[] args) {  
    System.out.println("Main method");  
    Test t1 = new Test();  
    Test t2 = new Test();  
}  
}
```

Output:

Static block
Main method
Non-static block
Constructor
Non-static block
Constructor

◆ Execution Flow (Important)

1. **Static block executes first** (only once)
2. Then **main() method runs**
3. When object is created:
 - a. Non-static block executes
 - b. Then constructor executes
4. For multiple objects:
 - a. Non-static block + constructor runs every time

◆ Rules of Blocks (Important Points)

Static Block Rules

- Executes **only once per class**
- Runs **before main()**
- Can access only **static variables**
- Cannot access non-static members directly

Non-Static Block Rules

- Executes **every time object is created**
- Runs **before constructor**
- Can access both static and non-static variables

General Rules

- Blocks are used for **initialization**
- Multiple static and non-static blocks allowed
- Static blocks execute in **top-to-bottom order**
- Non-static blocks also execute in **top-to-bottom order**

◆ Quick Summary

Block Type	Runs When	Runs How Many Times
Static Block	Class loading	Once
Non-static	Object creation	Every object